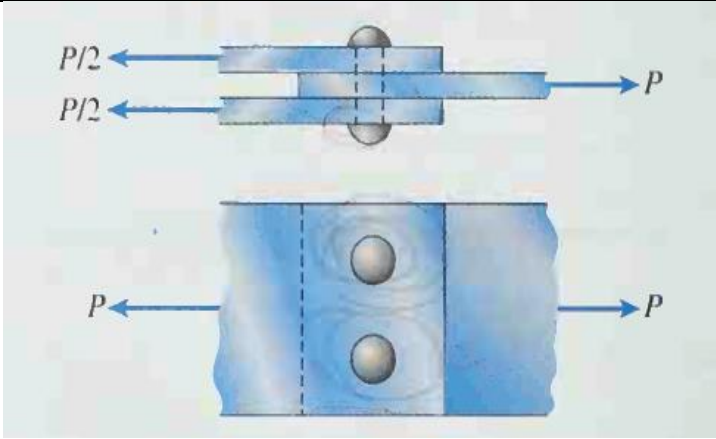
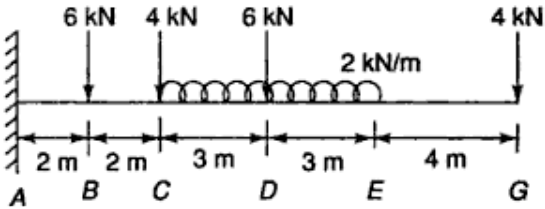
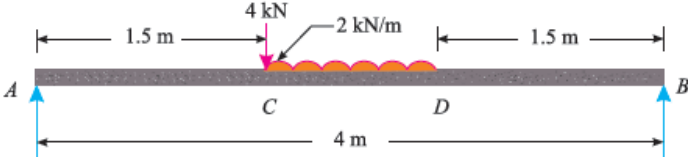


	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Semester Examination – Summer 2022 Course: B. Tech. Branch : Mechanical Engg. Semester : IV Subject Code & Name: BTMEC403 – Strength of Materials Max Marks: 60 Date: Duration: 3.00 Hrs.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Define and explain the term ‘Poisson’s ratio’. How is it calculated based on lateral and axial strains?	CO1	06
B)	A short hollow circular tube (with vertical axis) is applied with an axial load of 5 kN. The inner and outer diameters of the tube are 75 mm and 100 mm, respectively. The original length of the tube is 500 mm which is reduced by 0.2 mm after applying the load. Determine the compressive stress and strain in the tube. (Neglect the self weight of the tube and assume the tube does not buckle under the load).	CO2	06
C)	A stepped metal rod ABC (as shown in figure below) is subjected to an axial pull of 7 kN. The diameters of the two sections AB and section BC of the rod are 65 mm and 30 mm respectively. Find out in which of the two sections the stress will be maximum? Also find out the stresses in section AB and BC of the rod.	CO2	06
Q.2	Solve Any Two of the following.		
A)	Three steel plates of 18 mm thickness each are joined with the help of two rivets of 15 mm diameter each. If the plates are pulled (as shown in the figure below) with $P = 6$ kN, calculate the maximum shear stress induced in the rivets.	CO3	06

			
B)	The tensile stresses at a point across mutually perpendicular planes are 120 N/mm^2 and 60 N/mm^2 . Determine the normal, tangential and resultant stresses on a plane inclined at 30 degrees to the axis of minor stress.	CO3	06
C)	Define and explain following terms: A. Resilience B. Proof Resilience C. Modulus of Resilience	CO3	06
Q. 3 Solve Any Two of the following.			
A)	For the cantilever beam as shown in figure below, calculate the shear force and bending moment and draw the shear force and bending moment diagram. 	CO04	06
B)	A simply supported beam 4 m span is carrying loads as shown in figure. Draw shear force and bending moment diagrams for the beam. 	CO4	06
C)	Describe various types of supports for beam with the help of neat sketch.	CO4	06
Q.4 Solve Any Two of the following.			
A)	A wooden beam 100 mm wide, 250 mm deep and 3 m long is carrying a uniformly distributed load of 40 kN/m . Determine the maximum shear stress		06

	and sketch the variation of shear stress and along the depth of the beam.		
B)	Explain the assumptions of theory of simple bending.		06
C)	A beam 6 m long, simply supported at its ends, is carrying a point load of 50 kN at its center. The moment of inertia of the beam is given as equal to $78 \times 10^6 \text{ mm}^4$. If E for the material of the beam is $2.1 \times 10^5 \text{ mm}^4$. Calculate: 1. Deflection at the center of the beam and 2. Slope at the support.	CO5	06
Q. 5	Solve Any Two of the following.		
A)	Find the maximum shear stress induced in a solid circular shaft of diameter 15 cm when the shaft transmits 150 kW power at 180 rpm.	CO5	06
B)	Describe various types of end conditions of columns.	CO6	06
C)	Write short note on 'Limitation of Euler's formula'.	CO6	06
	*** End ***		